

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIP506C - APPLIED HYDRAULICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The main purpose of a spillway in a dam is to:
A) Increase water storage capacity
B) Reduce sedimentation in the reservoir
C) Generate electricity only
D) Release excess water safely during floods
 - 2 A stilling basin in a spillway structure is primarily used for:
A) Dissipating energy of flowing water
B) Storing additional water
C) Measuring flow velocity
D) Irrigation channeling
 - 3 The type of spillway commonly used for arch and gravity dams is:
A) Shaft spillway
B) Overflow spillway
C) Side channel spillway
D) Ogee spillway
 - 4 The Darcy–Weisbach equation is used to calculate:
A) Water surface elevation
B) Spillway discharge
C) Friction losses in pipe flow
D) Pump head
 - 5 The Colebrook–White equation is generally applicable to:
A) Only laminar flow
B) Turbulent flow in rough pipes
C) Open channel flows
D) Steady uniform flow

- 6 The line representing total mechanical energy per unit weight in a pipe system is called:
 - A) Water table line
 - B) Hydraulic Grade Line (HGL)
 - C) Pump curve
 - D) Energy Grade Line (EGL)
- 7 Boundary layer separation occurs when:
 - A) Flow velocity increases
 - B) Adverse pressure gradient forces flow to detach from surface
 - C) Water temperature decreases
 - D) Surface becomes frictionless
- 8 Stokes' law is applicable for calculating drag on:
 - A) High Reynolds number flow
 - B) Large submerged objects
 - C) Small spherical particles in creeping flow
 - D) Flow over flat plates only
- 9 When a jet strikes a flat plate normally (at right angle), the force exerted is proportional to:
 - A) Velocity only
 - B) Density only
 - C) Momentum change of the jet
 - D) Viscosity of water
- 10 Moving vanes compared to stationary vanes require:
 - A) Less force and provide higher efficiency
 - B) More energy input
 - C) Same force
 - D) No momentum change
- 11 Cavitation in pumps is mainly caused by:
 - A) Excess lubrication
 - B) Vapour bubble formation due to low pressure
 - C) Low fluid viscosity
 - D) High water temperature only
- 12 Pump efficiency decreases significantly when:
 - A) Pump is installed near water level
 - B) Suction lift is high
 - C) Discharge pipe length is reduced
 - D) Cavitation is avoided

Q2. Solve the following.

- A) Describe the major features of dams based on their type and design basis. Explain with examples. <https://www.batuonline.com> 6
- B) Explain the different types of spillways used in flood control and discuss the design considerations for a suitable spillway. 6

Q3. Solve the following.

- A) Explain Darcy–Weisbach and Hazen–Williams equations used for estimating friction losses in pipe flow. 6
- B) Discuss the role of energy line and hydraulic grade line in the analysis of pipe networks. 6

- Q4. Solve Any Two of the following.
- A) Define boundary layer. Explain its growth over a flat plate and differentiate laminar and turbulent boundary layers. 6
 - B) Define the following boundary layer thicknesses with sketches: 1) Displacement thickness, 2) Momentum thickness, 3) Energy thickness 6
 - C) Explain drag and lift on submerged bodies. Discuss the behavior for plates, cylinders, and aerofoils. 6
- Q5. Solve Any Two of the following.
- A) Explain the working principle of jet impact on a stationary curved vane with a velocity diagram. 6
 - B) Derive an expression for the force exerted by a jet of water on a moving flat plate. 6
 - C) Explain impulse–momentum principle with reference to jet impact on a stationary flat plate. 6
- Q6. Solve Any Two of the following.
- A) Explain the importance of pump location and its effect on pump performance. 6
 - B) Explain the pump performance curve and its significance. 6
 - C) Discuss the performance of a multi-pump system and list factors influencing its hydraulic behavior. 6

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